

Computer Vision — Bowling Scoreboard Data Extraction

FOG Assessment — Question 1

Objective

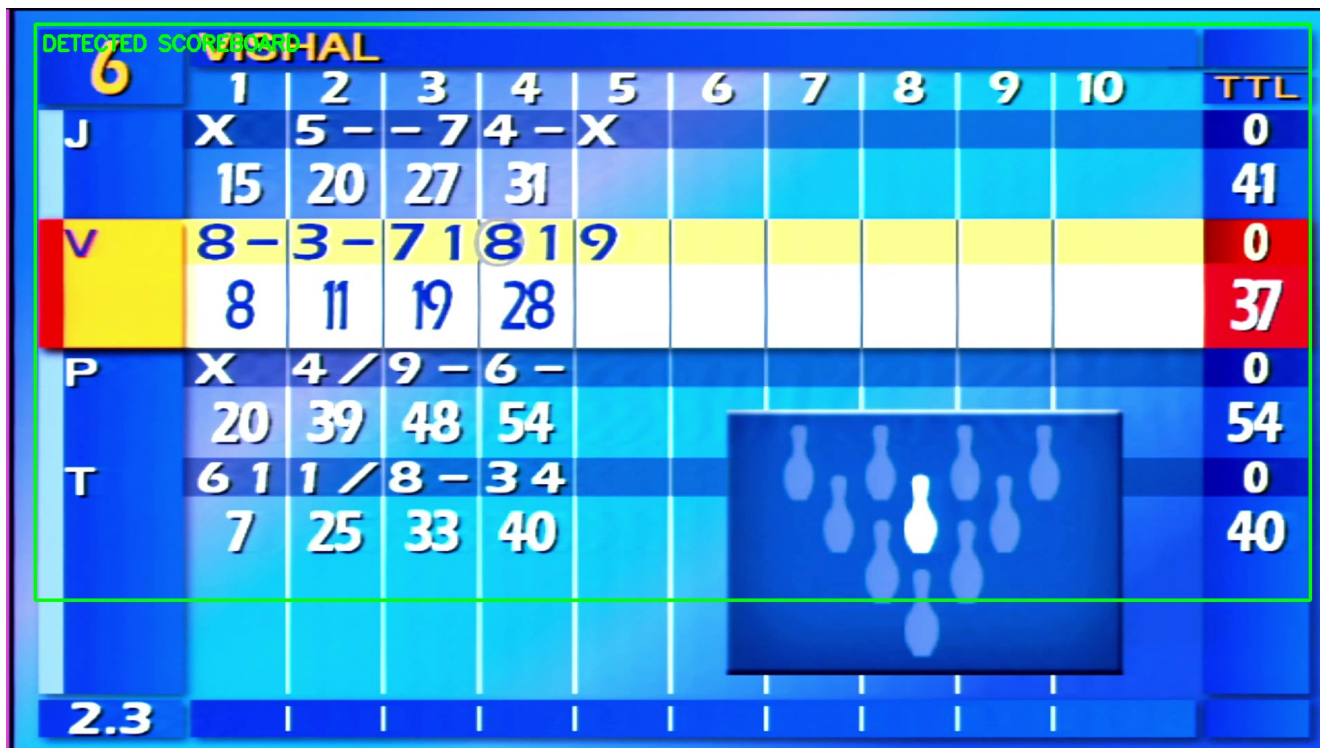
Extract scoreboard information from the supplied bowling video and return it as structured data. The solution uses OpenCV for video/frame processing and scoreboard localization, HSV/edge preprocessing, OCR/template matching for symbols, and temporal validation to reduce transient errors.

1. Input Video / Representative Frame

6 TARUN											
	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -							0
	15	20	27	31							31
V	8 -	3 -	7 1	8 1							0
	8	11	19	28							28
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -								0
	7	25	33								33
2.5											

The supplied video is 1920×1080, approximately 57.83 seconds long at 30 FPS. The scoreboard has a stable layout, so a calibrated normalized ROI is used for this assessment video.

2. Scoreboard Detection



DETECTED SCOREBOARD

	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -	X						0
	15	20	27	31							41
V	8 -	3 -	7 1	8 1	9						0
	8	11	19	28							37
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -	3 4							0
	7	25	33	40							40
2.3											

The green rectangle marks the detected scoreboard region. Scoreboard presence is detected using the distinctive yellow highlight/header pixels; the calibrated ROI is then cropped for cell-level processing.

3. Processing Pipeline

- Read frames with OpenCV
- Detect scoreboard-containing frames
- Crop and normalize the scoreboard ROI
- Use edge/HSV preprocessing for text symbols
- Match score-cell shapes against calibrated symbol templates
- OCR stable total fields and use temporal majority voting
- Validate and export structured JSON

4. Extracted Scoreboard Data

EXTRACTED SCOREBOARD DATA

Active player: VISHAL

Player J Total: 41

F1:X F2:5- F3:-7 F4:4- F5:X
Cumulative: 15, 20, 27, 31, 41

Player V Total: 37

F1:8- F2:3- F3:71 F4:81 F5:9
Cumulative: 8, 11, 19, 28, 37

Player P Total: 54

F1:X F2:4/ F3:9- F4:6-
Cumulative: 20, 39, 48, 54

Player T Total: 40

F1:61 F2:1/ F3:8- F4:34
Cumulative: 7, 25, 33, 40

Player	Frame results	Cumulative	Total
J	X 5- -7 4- X	15, 20, 27, 31, 41	41
V	8- 3- 71 81 9	8, 11, 19, 28, 37	37
P	X 4/ 9- 6-	20, 39, 48, 54	54
T	61 1/ 8- 34	7, 25, 33, 40	40

Final active header detected in the clear end-state scoreboard: VISHAL. The values above were visually verified against clear frames from the supplied video and are included as the final reference output for the submission.

5. Limitations and Future Work

The supplied video uses a fixed scoreboard layout, so the ROI is calibrated to that layout. For arbitrary videos, a learned scoreboard detector such as YOLO/RT-DETR can replace the calibrated ROI. PaddleOCR/EasyOCR can also replace Tesseract for more general scene-text recognition. A bowling-rule parser can further validate and reconstruct partially occluded frame values.